# PnD service final (concise)

# Planning & Discovery Service - Technical Specification v4.0

**MetaOrcha Orchestration System**
**Version:** 4.0 (Production-Ready)
**Date:** February 2026
**Status:** ✅ Ready for Implementation

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## Table of Contents



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## 1. Executive Summary

### 1.1 Service Purpose

The Planning & Discovery Service (P&D Service) is a **dual-function orchestration component** in the MetaOrcha ecosystem:

**Function 1: Agent Manifest Processing**

- Consumes agent registration events from Registry Service via Kafka
- Transforms structured JSON manifests into semantic strings using TDWA template
- Generates vector embeddings (1536-dim) using text-embedding-3-small
- Stores in PostgreSQL with pgvector for fast similarity search

**Function 2: Query Planning & Optimization**

- Transforms natural language queries into validated, executable workflows
- Uses optimized 3-stage pipeline with **80-95% LLM call reduction**
- Achieves **sub-200ms agent resolution** through PostgreSQL indices
- Provides **100% pre-flight validation** for workflow reliability

### 1.2 Key Performance Targets

```
┌──────────────────────────────────────────────────────────────┐
│                    PERFORMANCE TARGETS                       │
├──────────────────────────────────────────────────────────────┤
│                                                              │
│  Metric                    │  Target          │  vs Legacy   │
│  ──────────────────────────┼──────────────────┼──────────────│
│  Query Planning Latency    │  2-3 seconds     │  10× faster  │
│  Agent Resolution          │  <200ms          │  No LLM      │
│  LLM Calls per Query       │  1-2 calls       │  -95%        │
│  Cost per Query            │  $0.0001         │  -99%        │
│  Agent Registry Scale      │  100K+ agents    │  Proven      │
│  Concurrent Queries        │  1000+ QPS       │  Scalable    │
└──────────────────────────────────────────────────────────────┘
```

### 1.3 Technology Stack

| Component | Technology | Version | Purpose |
|---|---|---|---|
| **Database** | PostgreSQL | 16+ | Primary data store |
| **Vector Search** | pgvector | 0.7.0+ | HNSW indices |
| **Message Queue** | Apache Kafka | 3.6+ | Event streaming |
| **LLM Provider** | OpenRouter | Latest | API gateway |
| **Primary LLM** | Claude 3.7 Sonnet | Latest | Decomposition |
| **Fast LLM** | Gemini 1.5 Flash | Latest | Resolution |
| **Embeddings** | text-embedding-3-small | Latest | 1536-dim vectors |
| **Language** | Python | 3.11+ | Service implementation |
| **Framework** | FastAPI | 0.109+ | REST API |
| **ORM** | Prisma | 0.11+ | Database access |
| **Package Manager** | uv | Latest | Monorepo tooling |

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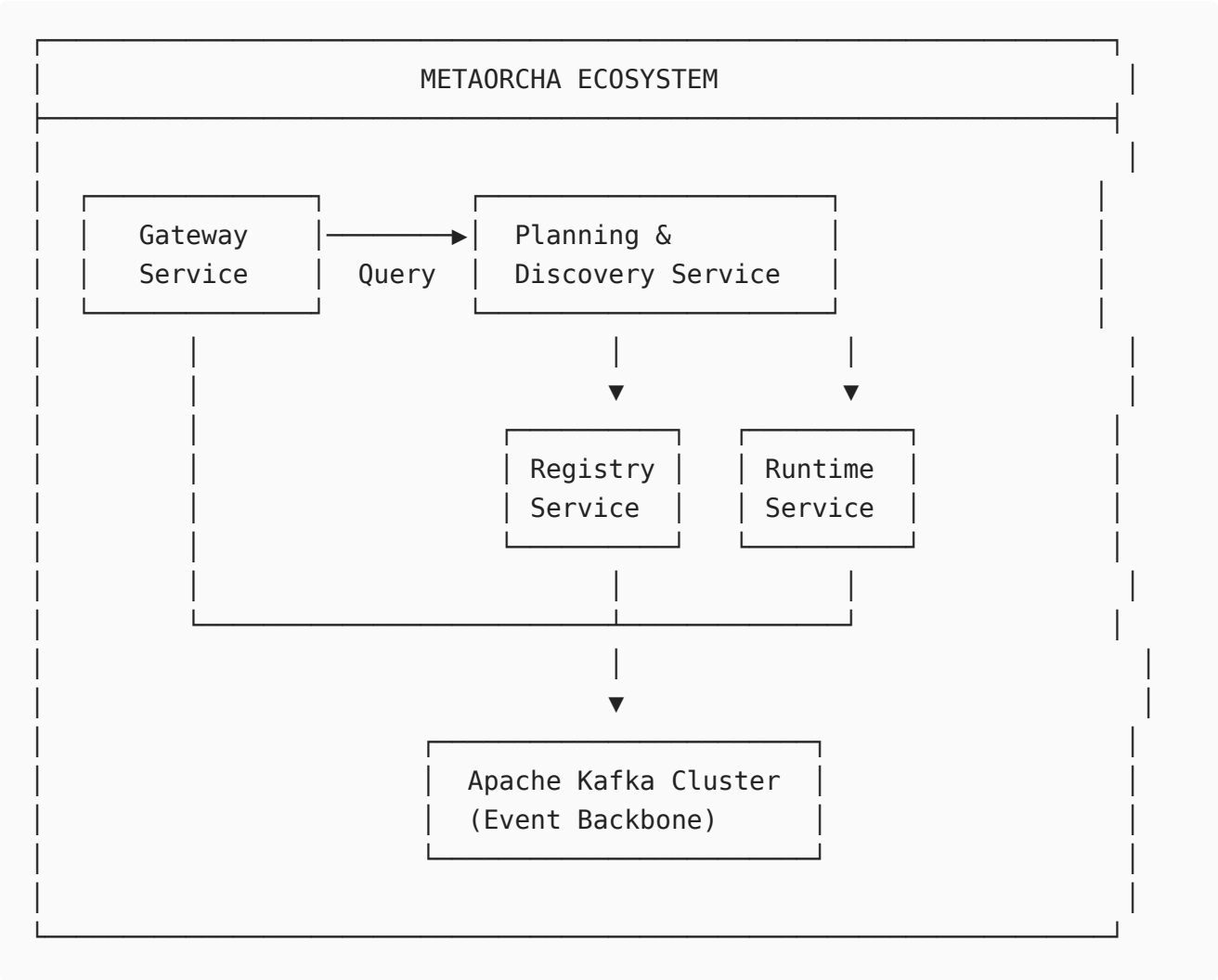
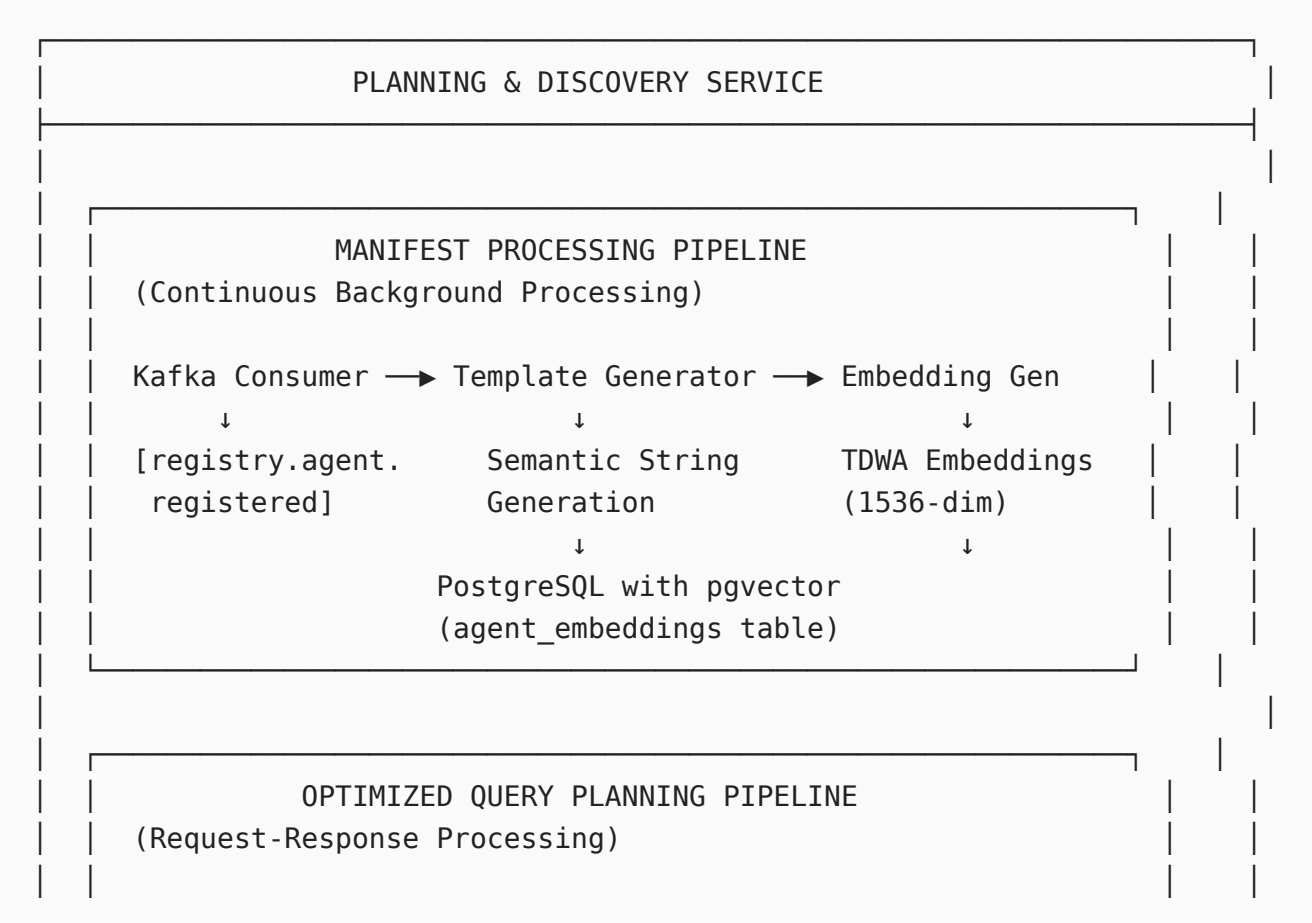
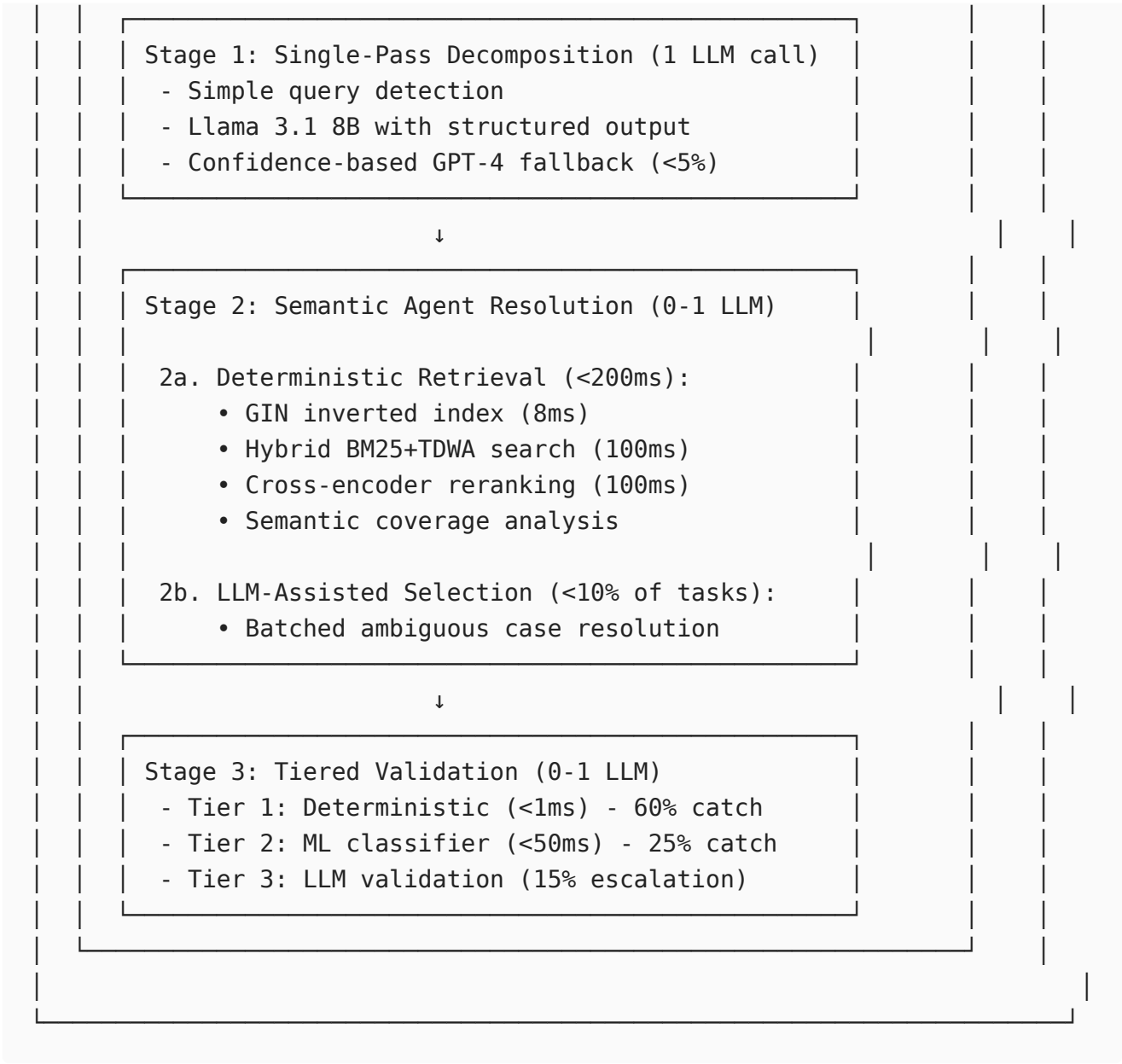
## 2. Architecture Overview

### 2.1 High-Level System Context




## 2.2 Service Internal Architecture



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## 3. Infrastructure Dependencies

### 3.1 PostgreSQL 16+ with Extensions

**Required Extensions:**

```
-- Core vector search
CREATE EXTENSION IF NOT EXISTS vector;

-- Trigram similarity for fuzzy matching
CREATE EXTENSION IF NOT EXISTS pg_trgm;

-- Better GIN index performance
CREATE EXTENSION IF NOT EXISTS btree_gin;
```

```
-- Query performance monitoring
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
```

**Connection Configuration:**

```
# config/database.py

DATABASE_CONFIG = {
    "dsn": os.getenv("DATABASE_URL"),
    "min_size": 10,
    "max_size": 20,
    "max_queries": 50000,
    "max_inactive_connection_lifetime": 300,
    "timeout": 60,
    "command_timeout": 30,
    "server_settings": {
        "jit": "off",  # Disable JIT for vector operations
        "work_mem": "256MB",  # Larger for complex queries
        "maintenance_work_mem": "512MB"  # For index builds
    }
}
```

## 3.2 Apache Kafka 3.6+

**Topics:**

| Topic | Purpose | Partitions | Retention |
|---|---|---|---|
| `registry.agent.registered` | Agent manifest events | 6 | 7 days |
| `gateway.user.query` | User query requests | 12 | 24 hours |
| `planning.manifest.created` | Completed workflow plans | 6 | 30 days |
| `planning.validation.failed` | Failed plan attempts | 3 | 7 days |
| `planning.metrics` | Telemetry events | 3 | 3 days |

**Consumer Groups:**

```
KAFKA_CONSUMERS = {
    "manifest_processor": {
        "group_id": "planning-manifest-processors",
        "topics": ["registry.agent.registered"],
        "auto_offset_reset": "earliest",
        "enable_auto_commit": True,
        "max_poll_records": 50
    },
    "query_handler": {
```

```
        "group_id": "planning-query-handlers",
        "topics": ["gateway.user.query"],
        "auto_offset_reset": "latest",
        "enable_auto_commit": False,  # Manual commit after processing
        "max_poll_records": 10
    }
}
```

## 3.3 LLM Provider (OpenRouter)

**Model Configuration:**

```
# config/llm.py

LLM_MODELS = {
    "decomposition": {
        "primary": "meta-llama/llama-3.1-8b-instruct",
        "fallback": "openai/gpt-4o"
    },
    "resolution": {
        "primary": "google/gemini-1.5-flash"
    },
    "reranking": {
        "primary": "anthropic/claude-3-7-sonnet"
    },
    "validation": {
        "primary": "openai/gpt-4o-mini"
    },
    "embeddings": {
        "primary": "openai/text-embedding-3-small"
    }
}

OPENROUTER_CONFIG = {
    "base_url": "https://openrouter.ai/api/v1",
    "timeout": 60,
    "max_retries": 3,
    "retry_delay": 1.0,
    "prompt_caching": True  # Enable for system prompts
}
```

## 3.4 Python Dependencies (uv)

**pyproject.toml:**

```
[project]
name = "planning-discovery-service"
version = "4.0.0"
```

```toml
requires-python = ">=3.11"

dependencies = [
    # Core Framework
    "fastapi>=0.109.0",
    "uvicorn[standard]>=0.27.0",
    "pydantic>=2.6.0",
    "pydantic-settings>=2.1.0",

    # Database
    "asyncpg>=0.29.0",
    "psycopg[binary]>=3.1.0",
    "prisma>=0.11.0",

    # Message Queue
    "aiokafka>=0.10.0",

    # LLM & Embeddings
    "openai>=1.12.0",
    "anthropic>=0.18.0",
    "httpx>=0.26.0",
    "tenacity>=8.2.0",

    # ML & NLP
    "sentence-transformers>=2.3.1",
    "torch>=2.1.0",
    "scikit-learn>=1.4.0",
    "joblib>=1.3.0",
    "numpy>=1.26.0",
    "nltk>=3.8.1",

    # Monitoring
    "prometheus-client>=0.19.0",
    "structlog>=24.1.0",

    # Utilities
    "python-dotenv>=1.0.0",
    "redis>=5.0.0",
]

[project.optional-dependencies]
dev = [
    "pytest>=7.4.0",
    "pytest-asyncio>=0.21.0",
    "pytest-cov>=4.1.0",
    "pytest-mock>=3.12.0",
    "ruff>=0.1.0",
    "mypy>=1.8.0",
]
```

**Installation:**

```
# Using uv in monorepo
cd services/planning-discovery
uv pip install -e .
uv pip install -e ".[dev]"
```

---

# 4. Database Schema & Indices

## 4.1 Complete Prisma Schema

```
// schema.prisma

generator client {
  provider             = "prisma-client-py"
  interface            = "asyncio"
  recursive_type_depth = 5
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

//
==========================================================================
==
// AGENT REGISTRY (Shared with Registry Service)
//
==========================================================================
==

model Agent {
  id               String                          @id @default(cuid())
  did              String                          @unique
  name             String
  version          String
  description      String?                         @db.Text
  tags             String[]
  protocol_type    ProtocolType
  health_status    HealthStatus                    @default(UNKNOWN)
  last_heartbeat   DateTime?
  is_active        Boolean                         @default(true)
  created_at       DateTime                        @default(now())
  updated_at       DateTime                        @updatedAt
```

```
  // Relations
  transport         Transport?
  security          SecurityConfig?
  payment           PaymentConfig?
  capabilities      Capability[]
  embedding         AgentEmbedding?

  @@index([protocol_type])
  @@index([health_status])
  @@index([is_active])
  @@map("agents")
}

model Transport {
  id           String        @id @default(cuid())
  agent_id     String        @unique
  type         TransportType
  http_url     String?
  sse_url      String?
  stdio_config Json?

  agent        Agent         @relation(fields: [agent_id], references:
[id], onDelete: Cascade)

  @@map("transports")
}

model SecurityConfig {
  id              String              @id @default(cuid())
  agent_id        String              @unique
  auth_strategy   AuthStrategy
  api_key         String?
  oauth_config    Json?
  mtls_config     Json?

  agent           Agent               @relation(fields: [agent_id],
references: [id], onDelete: Cascade)

  @@map("security_configs")
}

model PaymentConfig {
  id         String  @id @default(cuid())
  agent_id   String  @unique
  enabled    Boolean @default(false)
  chain_id   String?
  asset      String?
  per_call   Decimal?
```

```
  agent      Agent   @relation(fields: [agent_id], references: [id],
onDelete: Cascade)

  @@map("payment_configs")
}

model Capability {
  id              String       @id @default(cuid())
  agent_id        String
  type            CapabilityType
  capability_id   String
  name            String
  description     String?      @db.Text
  input_schema    Json
  output_schema   Json

  agent           Agent        @relation(fields: [agent_id], references:
[id], onDelete: Cascade)

  @@index([agent_id])
  @@index([type])
  @@map("capabilities")
}

//
================================================================================
==
// VECTOR SEARCH (Planning & Discovery Service)
//
================================================================================
==

model AgentEmbedding {
  id                String                        @id @default(cuid())
  agent_id          String                        @unique

  // Semantic string representation
  templated_string  String                        @db.Text

  // Vector embedding (1536 dimensions)
  embedding         Unsupported("vector(1536)")?

  // Component embeddings for TDWA
  name_embedding            Unsupported("vector(1536)")?
  description_embedding     Unsupported("vector(1536)")?
  capabilities_embedding    Unsupported("vector(1536)")?
  tags_embedding            Unsupported("vector(1536)")?

  created_at        DateTime                      @default(now())
```

```
  updated_at        DateTime                      @updatedAt

  // Relations
  agent             Agent                         @relation(fields:
[agent_id], references: [id], onDelete: Cascade)

  @@index([agent_id])
  @@map("agent_embeddings")
}

//
================================================================================
==
// PLAN EXECUTION HISTORY (for ML Classifier Training)
//
================================================================================
==

model PlanExecution {
  id                      String    @id @default(cuid())
  plan_hash               String
  workflow_manifest       Json

  // Execution results
  success                 Boolean
  error_message           String?   @db.Text
  execution_time_ms       Int?

  // Features for ML model
  node_count              Int
  edge_count              Int
  max_depth               Int
  parallel_width          Int
  agent_reliability_mean  Float
  agent_reliability_min   Float
  has_loops               Boolean
  has_conditionals        Boolean
  conditional_depth       Int

  created_at              DateTime  @default(now())

  @@index([plan_hash])
  @@index([success])
  @@index([created_at])
  @@map("plan_executions")
}

//
================================================================================
```

```
==
// ENUMS
//
================================================================================
==

enum ProtocolType {
  MCP
  A2A
  ACP
}

enum TransportType {
  HTTP
  SSE
  STDIO
}

enum AuthStrategy {
  API_KEY
  OAUTH2
  MTLS
  NONE
}

enum HealthStatus {
  HEALTHY
  UNHEALTHY
  UNKNOWN
}

enum CapabilityType {
  TOOL
  RESOURCE
  PROMPT
}
```

## 4.2 PostgreSQL Index Creation Scripts

```
-- scripts/db/001_create_vector_indices.sql

-- Enable extensions
CREATE EXTENSION IF NOT EXISTS vector;
CREATE EXTENSION IF NOT EXISTS pg_trgm;
CREATE EXTENSION IF NOT EXISTS btree_gin;
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;

--
================================================================================
```

```
==
-- VECTOR SEARCH INDICES (HNSW for fast similarity search)
--
================================================================================
==

-- Primary combined embedding (TDWA weighted average)
CREATE INDEX IF NOT EXISTS agent_embeddings_combined_hnsw_idx
ON agent_embeddings
USING hnsw (embedding vector_cosine_ops)
WITH (m = 16, ef_construction = 64);

-- Component embeddings (optional, for debugging)
CREATE INDEX IF NOT EXISTS agent_embeddings_name_hnsw_idx
ON agent_embeddings
USING hnsw (name_embedding vector_cosine_ops)
WITH (m = 8, ef_construction = 32);

--
================================================================================
==
-- FULL-TEXT SEARCH (GIN on tsvector)
--
================================================================================
==

-- Add tsvector column to agents table
ALTER TABLE agents
ADD COLUMN IF NOT EXISTS search_vector tsvector
GENERATED ALWAYS AS (
    setweight(to_tsvector('english', coalesce(name, '')), 'A') ||
    setweight(to_tsvector('english', coalesce(description, '')), 'B') ||
    setweight(to_tsvector('english', coalesce(array_to_string(tags, ' '),
'')), 'C')
) STORED;

-- GIN index on tsvector for fast full-text search
CREATE INDEX IF NOT EXISTS agents_search_vector_idx
ON agents
USING GIN (search_vector);

--
================================================================================
==
-- INVERTED INDICES (GIN for array overlap)
--
================================================================================
==
```

```
-- Tags array index (for keyword filtering)
CREATE INDEX IF NOT EXISTS agents_tags_gin_idx
ON agents
USING GIN (tags);

--
================================================================================
==
-- STANDARD B-TREE INDICES
--
================================================================================
==

-- Protocol type filtering
CREATE INDEX IF NOT EXISTS agents_protocol_type_idx
ON agents (protocol_type)
WHERE is_active = true;

-- Health status filtering
CREATE INDEX IF NOT EXISTS agents_health_status_idx
ON agents (health_status)
WHERE is_active = true;

-- Composite index for common filters
CREATE INDEX IF NOT EXISTS agents_active_healthy_idx
ON agents (is_active, health_status, protocol_type);

-- Capability type filtering
CREATE INDEX IF NOT EXISTS capabilities_type_agent_idx
ON capabilities (type, agent_id);
```

### 4.3 Database Initialization Script

```
# scripts/db/initialize_database.py

import asyncio
import asyncpg
from pathlib import Path
import os

async def initialize_database():
    """
    Initializes PostgreSQL database with all required extensions and
indices.
    Run this after Prisma migrations.
    """

    db_url = os.getenv("DATABASE_URL")
    conn = await asyncpg.connect(db_url)
```

```
    try:
        print("🔧 Initializing database...")

        # Read and execute SQL script
        sql_path = Path(__file__).parent / "001_create_vector_indices.sql"
        sql = sql_path.read_text()

        await conn.execute(sql)

        print("✓ Extensions enabled")
        print("✓ Vector indices created")
        print("✓ Full-text search configured")
        print("✓ Inverted indices created")

        # Verify indices
        indices = await conn.fetch("""
            SELECT indexname, tablename
            FROM pg_indexes
            WHERE schemaname = 'public'
            ORDER BY tablename, indexname
        """)

        print(f"\n📊 Created {len(indices)} indices:")
        for idx in indices:
            print(f"  • {idx['tablename']}.{idx['indexname']}")

        print("\n✅ Database initialization complete!")

    finally:
        await conn.close()

if __name__ == "__main__":
    asyncio.run(initialize_database())
```

### 4.4 Index Maintenance

```
-- scripts/db/maintain_indices.sql

-- Run weekly
VACUUM (ANALYZE, VERBOSE) agent_embeddings;
VACUUM ANALYZE agents;
VACUUM ANALYZE capabilities;

-- Run monthly or after major data changes
REINDEX INDEX CONCURRENTLY agent_embeddings_combined_hnsw_idx;
REINDEX INDEX CONCURRENTLY agents_search_vector_idx;
REINDEX INDEX CONCURRENTLY agents_tags_gin_idx;
```

```
-- Run daily
ANALYZE agent_embeddings;
ANALYZE agents;
ANALYZE capabilities;
```

---

# 5. Manifest Processing Pipeline

## 5.1 TDWA Semantic Template

**Template Structure:**

```
Agent: {name}.
Role: {description}.
Expertise: {tags_joined}.
Protocol: {protocol_type}.
Integrated Skills: {for each capability: {type} named {name} used for
{description} with inputs {input_params}}.
Networks: {supported_networks}.
Reliability: {success_rate}%.
```

**Example Output:**

```
Agent: CryptoOracle Pro.
Role: Real-time cryptocurrency price oracle for multi-chain environments.
Expertise: crypto, defi, price-feed, oracle, multi-chain.
Protocol: MCP.
Integrated Skills: TOOL named Get Crypto Price used for Fetches current
market price for any cryptocurrency symbol with inputs symbol, network,
currency; TOOL named Get Historical Prices used for Retrieves price
history for technical analysis with inputs symbol, timeframe, interval;
RESOURCE named Price Charts used for Access pre-generated price chart
images with inputs symbol, period.
Networks: Ethereum, Base, Polygon, Arbitrum, Optimism.
Reliability: 98.5%.
```

## 5.2 Template Generator Implementation

```python
# src/manifest_processing/template_generator.py

from typing import Dict, List

class SemanticTemplateGenerator:
    """
    Transforms agent manifests into TDWA-optimized semantic strings.
```

```
    TDWA (Tool Document Weighted Average) format emphasizes:
    - Agent name and description (high weight)
    - Capability names (high weight)
    - Tags and metadata (medium weight)
    - Detailed descriptions (lower weight)
    """

    def generate(self, manifest: Dict) -> str:
        """Generate semantic string from manifest."""

        # Extract core identity
        identity = manifest.get("identity", {})
        name = identity.get("name", "Unknown Agent")
        description = identity.get("description", "")
        tags = ", ".join(identity.get("tags", []))

        # Extract protocol info
        protocol = manifest.get("protocol", {})
        protocol_type = protocol.get("type", "UNKNOWN")

        # Build capabilities string
        capabilities = manifest.get("capabilities", [])
        skills = self._format_capabilities(capabilities)

        # Extract network support
        networks = self._extract_networks(manifest)
        networks_str = ", ".join(networks) if networks else "Not
specified"

        # Calculate reliability (if available)
        reliability = self._calculate_reliability(manifest)
        reliability_str = f"{reliability:.1f}%" if reliability else "Not
tracked"

        # Assemble template
        template = f"""Agent: {name}. \
Role: {description}. \
Expertise: {tags}. \
Protocol: {protocol_type}. \
Integrated Skills: {skills}. \
Networks: {networks_str}. \
Reliability: {reliability_str}."""

        return template

    def _format_capabilities(self, capabilities: List[Dict]) -> str:
        """Format capabilities list into semantic string"""

        formatted = []
```

```
        for cap in capabilities:
            cap_type = cap.get("type", "").upper()
            cap_name = cap.get("name", "")
            cap_desc = cap.get("description", "")

            # Extract input parameters
            input_schema = cap.get("input_schema", {})
            properties = input_schema.get("properties", {})
            input_params = ", ".join(properties.keys()) if properties else
"none"

            formatted.append(
                f"{cap_type} named {cap_name} used for {cap_desc} "
                f"with inputs {input_params}"
            )

        return "; ".join(formatted)

    def _extract_networks(self, manifest: Dict) -> List[str]:
        """Extract supported networks from manifest"""

        networks = set()
        capabilities = manifest.get("capabilities", [])

        for cap in capabilities:
            input_schema = cap.get("input_schema", {})
            properties = input_schema.get("properties", {})

            if "network" in properties:
                network_enum = properties["network"].get("enum", [])
                networks.update(network_enum)

            if "chain" in properties:
                chain_enum = properties["chain"].get("enum", [])
                networks.update(chain_enum)

        return sorted(list(networks))

    def _calculate_reliability(self, manifest: Dict) -> float:
        """Calculate reliability score if historical data available"""
        # Placeholder - would come from execution history
        return None
```

### 5.3 TDWA Embedding Generator

```
# src/manifest_processing/embedding_generator.py

from typing import Dict, List
```

```python
import numpy as np

class TDWAEmbeddingGenerator:
    """
    Generates TDWA (Tool Document Weighted Average) embeddings.
    Based on research from ScaleMCP paper.
    """

    # TDWA weights from research
    WEIGHTS = {
        "name": 0.30,
        "description": 0.25,
        "capability_names": 0.20,
        "tags": 0.15,
        "capability_descriptions": 0.10
    }

    def __init__(self, llm_provider):
        self.llm = llm_provider
        self.embedding_model = "text-embedding-3-small"
        self.embedding_dim = 1536
        self.max_tokens = 8000

    async def generate_tdwa_embedding(
        self,
        manifest: Dict
    ) -> Dict[str, List[float]]:
        """
        Generate TDWA embedding with all components.

        Returns:
            Dict with 'combined' embedding and component embeddings
        """

        identity = manifest.get("identity", {})
        capabilities = manifest.get("capabilities", [])

        # Extract components
        components = {
            "name": identity.get("name", ""),
            "description": identity.get("description", ""),
            "tags": " ".join(identity.get("tags", [])),
            "capability_names": " ".join([c.get("name", "") for c in
capabilities]),
            "capability_descriptions": " ".join([c.get("description", "")
for c in capabilities])
        }

        # Generate embeddings for each component
```

```
        embeddings = {}

        for component_name, text in components.items():
            if not text.strip():
                embeddings[component_name] = [0.0] * self.embedding_dim
            else:
                embeddings[component_name] = await self.llm.embed(
                    text=text,
                    model=self.embedding_model
                )

        # Calculate TDWA weighted average
        combined = np.zeros(self.embedding_dim)

        for component, weight in self.WEIGHTS.items():
            combined += weight * np.array(embeddings[component])

        embeddings["combined"] = combined.tolist()

        return embeddings
```

### 5.4 Kafka Consumer

```
# src/manifest_processing/consumer.py

from aiokafka import AIOKafkaConsumer
import json
import asyncio

class ManifestConsumer:
    """Kafka consumer for agent manifest events."""

    def __init__(self, kafka_brokers: List[str], llm_provider):
        self.consumer = AIOKafkaConsumer(
            "registry.agent.registered",
            bootstrap_servers=kafka_brokers,
            group_id="planning-manifest-processors",
            value_deserializer=lambda m: json.loads(m.decode('utf-8')),
            auto_offset_reset="earliest",
            enable_auto_commit=True
        )

        self.template_gen = SemanticTemplateGenerator()
        self.embedding_gen = TDWAEmbeddingGenerator(llm_provider)

    async def start(self):
        """Start consuming and processing manifests"""

        await self.consumer.start()
```

```
        try:
            async for message in self.consumer:
                try:
                    await self._process_manifest(message.value)
                except Exception as e:
                    logger.exception(f"Failed to process manifest: {e}")

        finally:
            await self.consumer.stop()

    async def _process_manifest(self, event: Dict):
        """Process single manifest event"""

        agent_id = event.get("agent_id")
        manifest = event.get("manifest")

        logger.info(f"Processing manifest for agent {agent_id}")

        # Generate semantic string
        semantic_string = self.template_gen.generate(manifest)

        # Generate TDWA embeddings
        embeddings = await
self.embedding_gen.generate_tdwa_embedding(manifest)

        # Store in database using raw SQL (pgvector requirement)
        await self._upsert_embedding(
            agent_id=agent_id,
            semantic_string=semantic_string,
            embeddings=embeddings
        )

        logger.info(f"✓ Stored embedding for {agent_id}")

    async def _upsert_embedding(
        self,
        agent_id: str,
        semantic_string: str,
        embeddings: Dict[str, List[float]]
    ):
        """Store embeddings using raw SQL"""

        query = """
            INSERT INTO agent_embeddings (
                id, agent_id, templated_string,
                embedding, name_embedding, description_embedding,
                capabilities_embedding, tags_embedding,
                created_at, updated_at
```

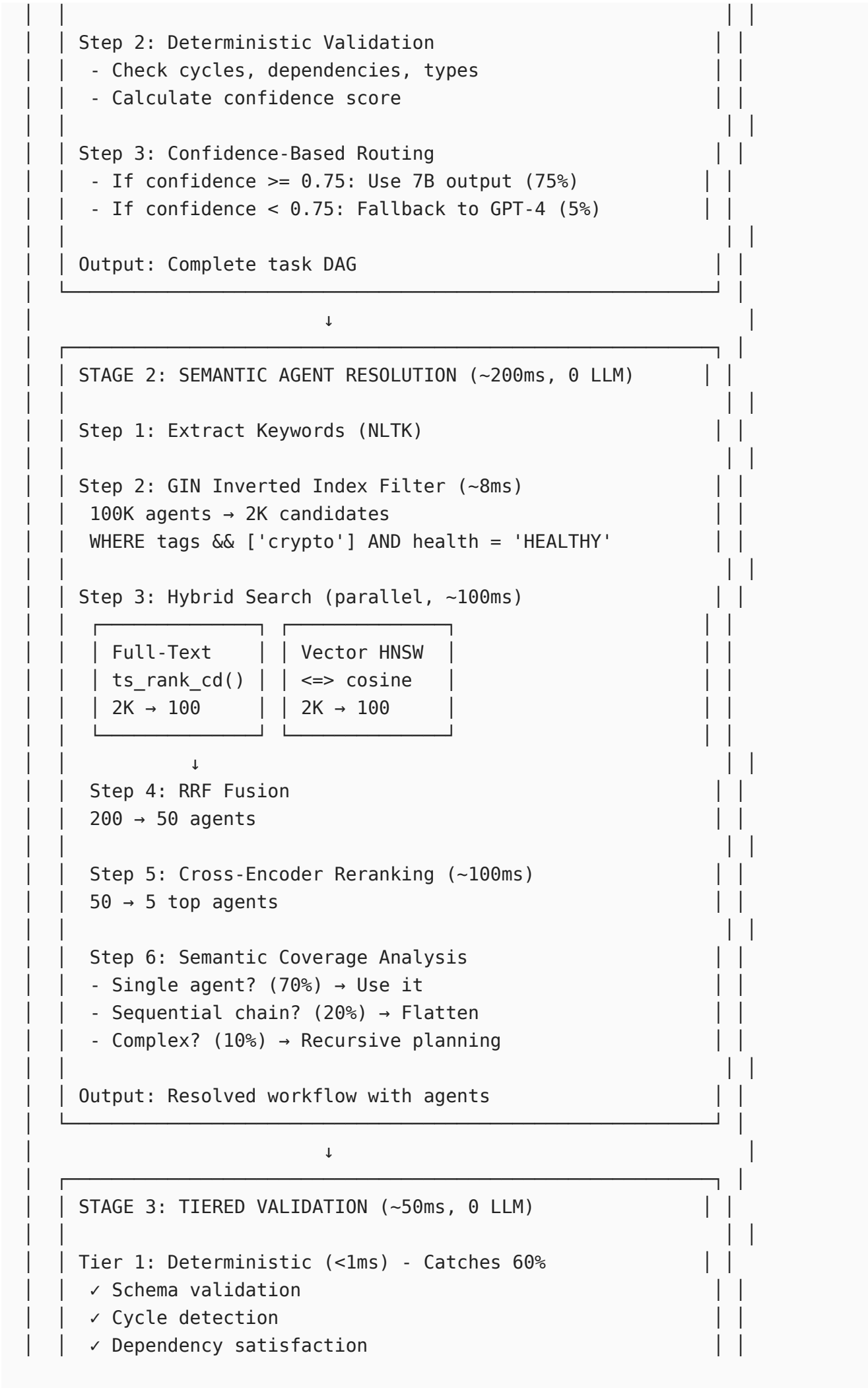
```
            )
            VALUES (
                gen_random_uuid(), $1, $2,
                $3::vector, $4::vector, $5::vector,
                $6::vector, $7::vector,
                NOW(), NOW()
            )
            ON CONFLICT (agent_id)
            DO UPDATE SET
                templated_string = $2,
                embedding = $3::vector,
                name_embedding = $4::vector,
                description_embedding = $5::vector,
                capabilities_embedding = $6::vector,
                tags_embedding = $7::vector,
                updated_at = NOW()
        """

        await db.execute_raw(
            query,
            agent_id,
            semantic_string,
            str(embeddings["combined"]),
            str(embeddings["name"]),
            str(embeddings["description"]),
            str(embeddings["capability_names"]),
            str(embeddings["tags"])
        )
```

---

## 6. Optimized Query Planning Pipeline

### 6.1 Complete Pipeline Flow

```
┌──────────────────────────────────────────────────────────────────────┐
│  User Query: "Monitor ETH price, alert if drops, trade if risk"      │
│                                                                      │
│  ┌──────────────────────────────────────────────────────────────┐    │
│  │ STAGE 1: SINGLE-PASS DECOMPOSITION (~1.5s, 1 LLM)            │    │
│  │                                                              │    │
│  │ Step 0: Pre-flight Check                                     │    │
│  │  - Detect simple queries (30-40%)                            │    │
│  │  - Skip decomposition if single-action                       │    │
│  │                                                              │    │
│  │ Step 1: Single LLM Call (Llama 3.1 8B)                       │    │
│  │  - Structured JSON output                                    │    │
│  │  - Complete DAG with dependencies                            │    │
```


Step 2: Deterministic Validation
- Check cycles, dependencies, types
- Calculate confidence score
Step 3: Confidence-Based Routing
- If confidence >= 0.75: Use 7B output (75%)
- If confidence < 0.75: Fallback to GPT-4 (5%)
Output: Complete task DAG
STAGE 2: SEMANTIC AGENT RESOLUTION (~200ms, 0 LLM)
Step 1: Extract Keywords (NLTK)
Step 2: GIN Inverted Index Filter (~8ms)
100K agents → 2K candidates
WHERE tags && ['crypto'] AND health = 'HEALTHY'
Step 3: Hybrid Search (parallel, ~100ms)
Full-Text
ts_rank_cd()
2K → 100
Vector HNSW
<=> cosine
2K → 100
Step 4: RRF Fusion
200 → 50 agents
Step 5: Cross-Encoder Reranking (~100ms)
50 → 5 top agents
Step 6: Semantic Coverage Analysis
- Single agent? (70%) → Use it
- Sequential chain? (20%) → Flatten
- Complex? (10%) → Recursive planning
Output: Resolved workflow with agents
STAGE 3: TIERED VALIDATION (~50ms, 0 LLM)
Tier 1: Deterministic (<1ms) - Catches 60%
✓ Schema validation
✓ Cycle detection
✓ Dependency satisfaction

```
│   │                                                              │ │
│   │ Tier 2: ML Classifier (<50ms) - Catches 25%                  │ │
│   │  ✓ Random Forest feasibility prediction                      │ │
│   │                                                              │ │
│   │ Tier 3: LLM Validation (15% escalation)                      │ │
│   │  ✓ Semantic compatibility check                              │ │
│   │                                                              │ │
│   │ Output: Validated workflow manifest                          │ │
│   └──────────────────────────────────────────────────────────────┘ │
│                                                                      │
│  Total: 2.15s | 1 LLM call | $0.0001                                │
└──────────────────────────────────────────────────────────────────────┘
```

### 6.2 Stage 1: Single-Pass Decomposition

```python
# src/planning/decomposition/single_pass_decomposer.py

from typing import Dict, List, Optional
from pydantic import BaseModel
import json

class DecompositionResult(BaseModel):
    """Result of decomposition"""
    success: bool
    tasks: List[Dict]
    edges: List[Dict]
    metadata: Dict
    confidence: float
    method: str  # 'skip', 'single_pass_7b', 'fallback_gpt4'
    llm_calls: int

class SinglePassDecomposer:
    """Decomposes queries into complete task DAGs in one LLM call."""

    def __init__(self, llm_provider):
        self.llm = llm_provider
        self.validator = DAGValidator()

        # Models
        self.small_model = "meta-llama/llama-3.1-8b-instruct"
        self.strong_model = "openai/gpt-4o"

        # Load cached system prompt
        self.system_prompt = self._load_system_prompt()

    async def decompose(
        self,
        user_query: str,
        context: Optional[Dict] = None
```

```
    ) -> DecompositionResult:
        """Main entry point for decomposition."""

        context = context or {}

        # Step 0: Pre-flight check
        if self._is_simple_query(user_query):
            return self._skip_decomposition(user_query)

        # Step 1: Single-pass decomposition
        decomposition = await self._single_pass_decompose(
            user_query,
            context
        )

        # Step 2: Deterministic validation
        validation = self.validator.validate(decomposition)

        # Step 3: Confidence-based routing
        if validation.confidence < 0.75:
            logger.warning(
                f"Low confidence ({validation.confidence:.2f}), "
                f"falling back to {self.strong_model}"
            )
            decomposition = await self._fallback_stronger_model(
                user_query,
                context,
                validation
            )

        return decomposition

    def _is_simple_query(self, query: str) -> bool:
        """Detect simple queries that don't need decomposition."""
        import re

        simple_patterns = [
            r"^(get|fetch|find|show|calculate|convert)\s+",
            r"^what (is|are) the\s+",
            r"^how much (is|are)\s+"
        ]

        complex_indicators = [
            r"\b(if|when|unless|after|then)\b",
            r"\b(every|repeatedly|until|while)\b",
            r"\band\s+(then|after)",
            r"\bor\b.*\belse\b"
        ]
```

```
        has_simple = any(re.search(p, query, re.I) for p in
simple_patterns)
        has_complex = any(re.search(p, query, re.I) for p in
complex_indicators)

        return has_simple and not has_complex

    def _skip_decomposition(self, query: str) -> DecompositionResult:
        """Create single-task decomposition for simple queries"""

        single_task = {
            "id": "task_1",
            "type": "agent_task",
            "description": query,
            "required_capabilities": self._infer_capabilities(query),
            "inputs": {},
            "depends_on": [],
            "expected_output_schema": {}
        }

        return DecompositionResult(
            success=True,
            tasks=[single_task],
            edges=[],
            metadata={"complexity": "simple", "estimated_steps": 1},
            confidence=0.95,
            method="skip",
            llm_calls=0
        )

    async def _single_pass_decompose(
        self,
        query: str,
        context: Dict
    ) -> DecompositionResult:
        """Generate complete DAG in single LLM call"""

        user_prompt = self._build_user_prompt(query, context)

        # Call small model with structured output
        response = await self.llm.complete(
            model=self.small_model,
            messages=[
                {"role": "system", "content": self.system_prompt},
                {"role": "user", "content": user_prompt}
            ],
            response_format={
                "type": "json_schema",
                "json_schema": TaskDAGSchema
```

```
            },
            temperature=0.3
        )

        dag = json.loads(response)

        return DecompositionResult(
            success=True,
            tasks=dag["tasks"],
            edges=dag["edges"],
            metadata=dag.get("metadata", {}),
            confidence=dag.get("metadata", {}).get("confidence", 0.8),
            method="single_pass_7b",
            llm_calls=1
        )

    def _load_system_prompt(self) -> str:
        """Load cached system prompt"""

        return """You are an expert workflow architect for multi-agent
systems.

TASK: Decompose user queries into complete task execution graphs (DAG) in
ONE response.

OUTPUT REQUIREMENTS:
1. Generate ALL tasks needed to fulfill the query
2. Specify task types: "agent_task", "router", "system_tool"
3. Define all dependencies explicitly
4. Annotate capability requirements per task
5. Include control flow (conditionals, loops)

CRITICAL RULES:
- Output ONLY valid JSON matching schema
- Each task must be independently executable
- No circular dependencies
- Router nodes handle ALL conditional logic
- System tools handle loops, waits, human approvals

TASK TYPES:
- agent_task: Requires external agent capability
- router: Conditional branching based on outputs
- system_tool: Built-in utilities (sleep, approval, loop_control)

OUTPUT JSON SCHEMA:
{
  "tasks": [
    {
      "id": "task_1",
```

```
      "type": "agent_task|router|system_tool",
      "description": "What this task does",
      "required_capabilities": {
        "capability_type": "TOOL|RESOURCE|PROMPT",
        "domains": ["domain1", "domain2"],
        "functions": ["func1", "func2"]
      },
      "inputs": {},
      "depends_on": ["task_id"],
      "expected_output_schema": {}
    }
  ],
  "edges": [
    {"from": "task_1", "to": "task_2", "condition": "optional"}
  ],
  "metadata": {
    "complexity": "simple|medium|high",
    "confidence": 0.0-1.0,
    "estimated_steps": 3
  }
}
"""
```

### 6.3 Stage 2: Hybrid Search Pipeline

```python
# src/planning/resolution/hybrid_search.py

from typing import List, Dict, Tuple
import asyncpg
import numpy as np
from sentence_transformers import CrossEncoder

class HybridSearchPipeline:
    """
    Complete hybrid search using PostgreSQL indices.
    """

    def __init__(self, pool: asyncpg.Pool, llm_provider):
        self.pool = pool
        self.llm = llm_provider
        self.cross_encoder = CrossEncoder(
            'cross-encoder/ms-marco-MiniLM-L-6-v2'
        )

    async def search(
        self,
        query: str,
        filters: Optional[Dict] = None,
        top_k: int = 5
```

```
    ) -> List[Dict]:
        """Execute complete hybrid search pipeline."""

        filters = filters or {}

        # Extract keywords
        keywords = self._extract_keywords(query)

        # Step 1: GIN inverted index filter (~8ms)
        candidate_ids = await self._inverted_index_filter(keywords,
filters)

        if not candidate_ids:
            return []

        # Step 2 & 3: Hybrid search in parallel (~100ms)
        fulltext_scores, vector_scores = await asyncio.gather(
            self._fulltext_search(query, candidate_ids),
            self._vector_search(query, candidate_ids)
        )

        # Step 4: Reciprocal Rank Fusion
        fused_results = self._reciprocal_rank_fusion(
            fulltext_scores,
            vector_scores,
            top_k=50
        )

        # Fetch full manifests
        agents = await self._fetch_agents(fused_results)

        # Step 5: Cross-encoder reranking (~100ms)
        top_agents = await self._cross_encoder_rerank(
            query,
            agents,
            top_k=top_k
        )

        return top_agents

    def _extract_keywords(self, query: str) -> List[str]:
        """Extract keywords using NLTK"""
        from nltk.tokenize import word_tokenize
        from nltk.corpus import stopwords

        tokens = word_tokenize(query.lower())
        stop_words = set(stopwords.words('english'))

        keywords = [
```

```
            t for t in tokens
            if t.isalpha() and len(t) > 2 and t not in stop_words
        ]

        return keywords

    async def _inverted_index_filter(
        self,
        keywords: List[str],
        filters: Dict
    ) -> List[str]:
        """Fast filtering using GIN indices"""

        sql = """
            SELECT id
            FROM agents
            WHERE
                health_status = 'HEALTHY'
                AND is_active = true
                AND ($1::text IS NULL OR protocol_type = $1::text)
                AND tags && $2::text[]
        """

        results = await self.pool.fetch(
            sql,
            filters.get("protocol_type"),
            keywords
        )

        return [r['id'] for r in results]

    async def _fulltext_search(
        self,
        query: str,
        candidate_ids: List[str]
    ) -> List[Tuple[str, float]]:
        """Full-text search using GIN on tsvector"""

        query_terms = ' & '.join(query.split())

        sql = """
            SELECT
                id,
                ts_rank_cd(search_vector, to_tsquery('english', $1), 32)
AS score
            FROM agents
            WHERE
                id = ANY($2::text[])
                AND search_vector @@ to_tsquery('english', $1)
```

```
            ORDER BY score DESC
            LIMIT 100
        """

        results = await self.pool.fetch(sql, query_terms, candidate_ids)

        return [(r['id'], float(r['score'])) for r in results]

    async def _vector_search(
        self,
        query: str,
        candidate_ids: List[str]
    ) -> List[Tuple[str, float]]:
        """Vector similarity search using HNSW"""

        # Generate query embedding
        query_emb = await self.llm.embed(query)

        sql = """
            SELECT
                agent_id,
                1 - (embedding <=> $1::vector) AS similarity
            FROM agent_embeddings
            WHERE agent_id = ANY($2::text[])
            ORDER BY embedding <=> $1::vector
            LIMIT 100
        """

        results = await self.pool.fetch(
            sql,
            str(query_emb),
            candidate_ids
        )

        return [(r['agent_id'], float(r['similarity'])) for r in results]

    def _reciprocal_rank_fusion(
        self,
        fulltext: List[Tuple[str, float]],
        vector: List[Tuple[str, float]],
        k: int = 60,
        top_k: int = 50
    ) -> List[str]:
        """Combine rankings using RRF"""
        from collections import defaultdict

        scores = defaultdict(float)

        # Add fulltext ranks
```

```
        for rank, (agent_id, _) in enumerate(fulltext, start=1):
            scores[agent_id] += 1 / (k + rank)

        # Add vector ranks
        for rank, (agent_id, _) in enumerate(vector, start=1):
            scores[agent_id] += 1 / (k + rank)

        # Sort by RRF score
        sorted_results = sorted(
            scores.items(),
            key=lambda x: x[1],
            reverse=True
        )

        return [agent_id for agent_id, _ in sorted_results[:top_k]]

    async def _cross_encoder_rerank(
        self,
        query: str,
        agents: List[Dict],
        top_k: int
    ) -> List[Dict]:
        """Rerank using cross-encoder"""

        # Create query-document pairs
        pairs = [
            [query, f"{a['name']}: {a['description']}"]
            for a in agents
        ]

        # Get scores
        scores = self.cross_encoder.predict(pairs, batch_size=32)

        # Combine with agents
        for agent, score in zip(agents, scores):
            agent['cross_encoder_score'] = float(score)
            agent['confidence'] = 'high' if score > 8.0 else 'medium' if
score > 6.0 else 'low'

        # Sort by score
        agents.sort(key=lambda x: x['cross_encoder_score'], reverse=True)

        return agents[:top_k]
```

### 6.4 Semantic Coverage Analysis

```
# src/planning/resolution/coverage_analyzer.py

from typing import Dict, List
```

```python
import numpy as np

class SemanticCoverageAnalyzer:
    """
    Analyzes if agents can fulfill task requirements.
    Uses semantic similarity instead of exact string matching.
    """

    def __init__(self, llm_provider, similarity_threshold: float = 0.75):
        self.llm = llm_provider
        self.threshold = similarity_threshold

    async def analyze_coverage(
        self,
        task: Dict,
        top_candidates: List[Dict]
    ) -> CoverageResult:
        """Determine orchestration strategy."""

        required_functions =
task['required_capabilities'].get('functions', [])

        if not required_functions:
            return CoverageResult(
                strategy='single_agent',
                agent=top_candidates[0],
                confidence='high',
                reasoning="No specific capability requirements"
            )

        # Embed required functions
        required_embeddings = {}
        for func in required_functions:
            required_embeddings[func] = await self.llm.embed(func)

        # Check each candidate
        for candidate in top_candidates:
            coverage = await self._calculate_coverage(
                required_functions,
                required_embeddings,
                candidate
            )

            if coverage['coverage_score'] >= 1.0:
                # Single agent covers everything
                return CoverageResult(
                    strategy='single_agent',
                    agent=candidate,
                    confidence='high',
```

```
                    coverage_details=coverage,
                    reasoning=f"Agent covers all {len(required_functions)}
required functions"
                )

        # Try to build sequential chain
        chain_result = await self._try_build_chain(
            required_functions,
            required_embeddings,
            top_candidates
        )

        if chain_result and len(chain_result) <= 3:
            return CoverageResult(
                strategy='sequential_chain',
                chain=chain_result,
                confidence='high',
                reasoning=f"Can chain {len(chain_result)} agents
sequentially"
            )

        # Need complex orchestration
        return CoverageResult(
            strategy='recursive_subgraph',
            available_agents=top_candidates,
            confidence='medium',
            reasoning="Requires complex multi-agent orchestration"
        )

    async def _calculate_coverage(
        self,
        required_functions: List[str],
        required_embeddings: Dict,
        agent: Dict
    ) -> Dict:
        """Calculate semantic coverage score"""

        matches = {}
        covered_count = 0

        # Get agent capabilities
        agent_capabilities = agent.get('capabilities', [])

        # Embed agent capabilities
        agent_cap_embeddings = {}
        for cap in agent_capabilities:
            text = f"{cap['name']}: {cap.get('description', '')}"
            agent_cap_embeddings[cap['name']] = await self.llm.embed(text)
```

```
        # Match each required function
        for req_func in required_functions:
            req_emb = required_embeddings[req_func]

            best_match = None
            best_similarity = 0.0

            for cap_name, cap_emb in agent_cap_embeddings.items():
                similarity = self._cosine_similarity(req_emb, cap_emb)

                if similarity > best_similarity:
                    best_similarity = similarity
                    best_match = cap_name

            # Check threshold
            if best_similarity >= self.threshold:
                matches[req_func] = {
                    'agent_capability': best_match,
                    'similarity': best_similarity,
                    'matched': True
                }
                covered_count += 1
            else:
                matches[req_func] = {
                    'agent_capability': best_match,
                    'similarity': best_similarity,
                    'matched': False
                }

        coverage_score = covered_count / len(required_functions)

        return {
            'coverage_score': coverage_score,
            'matches': matches,
            'covered_count': covered_count,
            'total_required': len(required_functions)
        }

    def _cosine_similarity(self, vec1: List[float], vec2: List[float]) ->
float:
        """Calculate cosine similarity"""
        vec1 = np.array(vec1)
        vec2 = np.array(vec2)

        dot_product = np.dot(vec1, vec2)
        norm1 = np.linalg.norm(vec1)
        norm2 = np.linalg.norm(vec2)

        if norm1 == 0 or norm2 == 0:
```

```
            return 0.0

        return float(dot_product / (norm1 * norm2))
```

### 6.5 Stage 3: Tiered Validation

```
# src/planning/validation/tiered_validator.py

from typing import Dict, List
from sklearn.ensemble import RandomForestClassifier
import joblib

class TieredValidator:
    """
    Three-tier progressive validation.

    Tier 1: Deterministic checks (60% catch rate)
    Tier 2: ML classifier (25% additional catch rate)
    Tier 3: LLM validation (15% of plans reach here)
    """

    def __init__(self, llm_provider):
        self.llm = llm_provider
        self.deterministic = DeterministicValidator()
        self.ml_classifier = self._load_ml_classifier()

    async def validate(self, workflow_dag: Dict) -> ValidationResult:
        """Progressive validation with escalation."""

        # Tier 1: Deterministic (always runs, <1ms)
        tier1_result = self.deterministic.validate(workflow_dag)

        if not tier1_result.is_valid:
            return tier1_result

        # Tier 2: ML classifier (<50ms)
        feasibility = self.ml_classifier.predict_feasibility(workflow_dag)

        if feasibility > 0.85:
            # High confidence - skip Tier 3
            return ValidationResult(
                is_valid=True,
                confidence=feasibility,
                tier="ml_classifier",
                llm_calls=0
            )
        elif feasibility < 0.60:
            # Low confidence - likely will fail
            return ValidationResult(
```

```
                is_valid=False,
                errors=["Low feasibility score from ML classifier"],
                confidence=feasibility,
                tier="ml_classifier",
                llm_calls=0
            )

        # Tier 3: LLM validation (uncertain cases only)
        tier3_result = await self._llm_validate(
            workflow_dag,
            tier1_result,
            feasibility
        )

        return tier3_result

    def _load_ml_classifier(self):
        """Load pre-trained feasibility classifier"""
        try:
            return joblib.load('models/feasibility_classifier.pkl')
        except FileNotFoundError:
            logger.warning("ML classifier not found, will skip Tier 2")
            return None

    async def _llm_validate(
        self,
        workflow_dag: Dict,
        tier1_result,
        feasibility: float
    ) -> ValidationResult:
        """LLM-based semantic validation"""

        prompt = f"""Validate this workflow plan for semantic correctness.

PLAN:
{json.dumps(workflow_dag, indent=2)}

CONCERNS:
- Structural validation: {tier1_result.warnings}
- ML feasibility score: {feasibility:.2f}

CHECK FOR:
1. Agent capability semantic compatibility
2. Domain-specific feasibility issues
3. Edge cases not caught by deterministic checks
4. Subtle dependency problems

Output JSON:
{{
```

```
  "is_valid": true/false,
  "issues": ["list", "of", "issues"],
  "confidence": 0.0-1.0,
  "suggestions": ["improvements"]
}}
"""

        response = await self.llm.complete(
            model="openai/gpt-4o-mini",
            messages=[{"role": "user", "content": prompt}],
            response_format={"type": "json_schema", "json_schema":
ValidationSchema}
        )

        result = json.loads(response)

        return ValidationResult(
            is_valid=result["is_valid"],
            errors=result["issues"],
            confidence=result["confidence"],
            suggestions=result.get("suggestions", []),
            tier="llm_semantic",
            llm_calls=1
        )
```

---

## 7. Standardized Data Schemas

### 7.1 Workflow Manifest Schema (Final)

```python
# src/schemas/workflow_manifest.py

from pydantic import BaseModel, Field
from typing import List, Dict, Optional, Literal, Union
from enum import Enum

class NodeType(str, Enum):
    """Node types in workflow DAG"""
    STANDARD = "standard"  # Concrete agent
    ROUTER = "router"  # Conditional logic
    SYSTEM_TOOL = "system_tool"  # Built-in utility

class CapabilityReference(BaseModel):
    """Reference to a specific agent capability"""
    capability_id: str
    type: Literal["TOOL", "RESOURCE", "PROMPT"]
    name: str
```

```
    input_schema: Dict
    output_schema: Dict

class TaskDescriptor(BaseModel):
    """Task description and requirements"""
    description: str
    inputs: Dict[str, any]
    expected_output: Optional[Dict] = None

class StandardNode(BaseModel):
    """Node representing a concrete agent invocation"""
    id: str
    type: Literal[NodeType.STANDARD] = NodeType.STANDARD
    agent_id: str  # DID of the agent
    capability: CapabilityReference
    task: TaskDescriptor
    depends_on: List[str] = []
    timeout_seconds: Optional[int] = 30
    retry_policy: Optional[Dict] = None

class RouterNode(BaseModel):
    """Node representing conditional branching"""
    id: str
    type: Literal[NodeType.ROUTER] = NodeType.ROUTER
    condition: str  # Expression to evaluate
    inputs: Dict[str, any]
    depends_on: List[str]
    routes: List[Dict[str, str]]  # [{condition, next_task, reason}]

class SystemToolNode(BaseModel):
    """Node for built-in system utilities"""
    id: str
    type: Literal[NodeType.SYSTEM_TOOL] = NodeType.SYSTEM_TOOL
    tool: Literal["sleep", "human_approval", "loop_control",
"set_variable"]
    inputs: Dict[str, any]
    depends_on: List[str]

class Edge(BaseModel):
    """Edge in workflow DAG"""
    from_: str = Field(alias="from")
    to: str
    condition: Optional[str] = None

class WorkflowManifest(BaseModel):
    """Complete workflow specification"""

    # Metadata
    id: str
```

```
    version: str = "4.0"
    created_at: str
    user_id: str

    # Graph structure
    nodes: List[Union[StandardNode, RouterNode, SystemToolNode]]
    edges: List[Edge]

    # Execution config
    max_concurrent_nodes: int = 10
    timeout_seconds: int = 300

    # Validation metadata
    validated: bool
    validation_tier: Literal["deterministic", "ml_classifier",
"llm_semantic"]
    confidence_score: float

    # Telemetry
    planning_metrics: Dict[str, any]

    class Config:
        allow_population_by_field_name = True
```

## 7.2 Internal Data Structures

```
# src/schemas/internal.py

from pydantic import BaseModel
from typing import List, Dict, Optional, Literal

class Task(BaseModel):
    """Intermediate task representation from decomposition"""
    id: str
    type: Literal["agent_task", "router", "system_tool"]
    description: str
    intent: Optional[str]
    required_capabilities: Dict
    inputs: Dict
    depends_on: List[str]
    expected_output_schema: Dict

class AgentSearchResult(BaseModel):
    """Agent candidate from search"""
    agent_id: str
    similarity_score: float
    cross_encoder_score: float
    confidence: Literal["high", "medium", "low"]
    manifest: Dict
```

```python
class CoverageResult(BaseModel):
    """Result of semantic coverage analysis"""
    strategy: Literal["single_agent", "sequential_chain",
"recursive_subgraph"]
    agent: Optional[AgentSearchResult]
    chain: Optional[List[AgentSearchResult]]
    available_agents: Optional[List[AgentSearchResult]]
    confidence: str
    coverage_details: Optional[Dict]
    reasoning: str

class ValidationResult(BaseModel):
    """Result of validation"""
    is_valid: bool
    errors: List[str] = []
    warnings: List[str] = []
    confidence: float
    tier: Literal["deterministic", "ml_classifier", "llm_semantic"]
    suggestions: List[str] = []
    llm_calls: int = 0
```

---

## 8. Service File Structure

```
services/planning-discovery/
├── README.md
├── pyproject.toml
├── .env.example
├── .gitignore
│
├── prisma/
│   ├── schema.prisma
│   └── migrations/
│
├── scripts/
│   ├── db/
│   │   ├── initialize_database.py
│   │   ├── 001_create_vector_indices.sql
│   │   └── maintain_indices.sql
│   ├── download_models.py
│   └── download_nltk_data.py
│
├── src/
│   ├── __init__.py
│   ├── main.py
│   ├── config.py
```

```
│   │
│   ├── api/
│   │   ├── __init__.py
│   │   ├── app.py
│   │   ├── routes/
│   │   │   ├── __init__.py
│   │   │   ├── health.py
│   │   │   └── planning.py
│   │   └── middleware/
│   │       ├── __init__.py
│   │       └── error_handler.py
│   │
│   ├── db/
│   │   ├── __init__.py
│   │   ├── prisma.py
│   │   └── connection_pool.py
│   │
│   ├── kafka/
│   │   ├── __init__.py
│   │   ├── consumer.py
│   │   ├── producer.py
│   │   └── topics.py
│   │
│   ├── llm/
│   │   ├── __init__.py
│   │   ├── provider.py
│   │   ├── openrouter.py
│   │   ├── ollama.py
│   │   └── config.py
│   │
│   ├── manifest_processing/
│   │   ├── __init__.py
│   │   ├── consumer.py
│   │   ├── template_generator.py
│   │   ├── embedding_generator.py
│   │   └── storage.py
│   │
│   ├── planning/
│   │   ├── __init__.py
│   │   ├── pipeline.py
│   │   │
│   │   ├── decomposition/
│   │   │   ├── __init__.py
│   │   │   ├── single_pass_decomposer.py
│   │   │   ├── schemas.py
│   │   │   ├── prompts.py
│   │   │   └── validator.py
│   │   │
│   │   ├── resolution/
```

```
│   │   │   ├── __init__.py
│   │   │   ├── hybrid_search.py
│   │   │   ├── coverage_analyzer.py
│   │   │   ├── cross_encoder.py
│   │   │   ├── recursive_resolver.py
│   │   │   └── keyword_extractor.py
│   │   │
│   │   └── validation/
│   │       ├── __init__.py
│   │       ├── tiered_validator.py
│   │       ├── deterministic.py
│   │       ├── ml_classifier.py
│   │       └── llm_validator.py
│   │
│   ├── schemas/
│   │   ├── __init__.py
│   │   ├── workflow_manifest.py
│   │   ├── internal.py
│   │   └── events.py
│   │
│   ├── monitoring/
│   │   ├── __init__.py
│   │   └── metrics.py
│   │
│   └── utils/
│       ├── __init__.py
│       ├── retry.py
│       ├── circuit_breaker.py
│       └── helpers.py
│
├── models/
│   └── feasibility_classifier.pkl
│
└── tests/
    ├── __init__.py
    ├── conftest.py
    ├── fixtures/
    │   ├── manifests.py
    │   └── queries.py
    │
    ├── unit/
    │   ├── test_template_generator.py
    │   ├── test_embedding_generator.py
    │   ├── test_decomposer.py
    │   ├── test_hybrid_search.py
    │   ├── test_coverage_analyzer.py
    │   └── test_validator.py
    │
    ├── integration/
```

```
│   ├── test_manifest_pipeline.py
│   ├── test_planning_pipeline.py
│   └── test_database.py
│
└── e2e/
    └── test_complete_workflow.py
```

---

## 9. Error Handling & Resilience

### 9.1 Error Categories

```python
# src/utils/errors.py

class PlanningError(Exception):
    """Base exception for planning errors"""
    pass

class UserError(PlanningError):
    """Recoverable errors caused by user input"""
    pass

class SystemError(PlanningError):
    """Retriable system errors"""
    pass

class FatalError(PlanningError):
    """Non-recoverable errors"""
    pass

# Specific error types
class AmbiguousQueryError(UserError):
    """Query is too ambiguous to decompose"""
    pass

class NoAgentsFoundError(UserError):
    """No agents match the requirements"""
    pass

class AgentUnavailableError(SystemError):
    """Agent exists but is unhealthy"""
    pass

class LLMTimeoutError(SystemError):
    """LLM request timed out"""
    pass
```

```python
class ValidationFailedError(FatalError):
    """Plan failed validation after max retries"""
    pass
```

### 9.2 Retry Logic with Tenacity

```python
# src/utils/retry.py

from tenacity import (
    retry,
    stop_after_attempt,
    wait_exponential,
    retry_if_exception_type
)

@retry(
    stop=stop_after_attempt(3),
    wait=wait_exponential(multiplier=1, min=1, max=10),
    retry=retry_if_exception_type(SystemError)
)
async def call_with_retry(func, *args, **kwargs):
    """Retry function calls on system errors"""
    return await func(*args, **kwargs)
```

### 9.3 Circuit Breaker

```python
# src/utils/circuit_breaker.py

from enum import Enum
import time

class CircuitState(Enum):
    CLOSED = "closed"  # Normal operation
    OPEN = "open"  # Failing, reject requests
    HALF_OPEN = "half_open"  # Testing recovery

class CircuitBreaker:
    """Circuit breaker for external dependencies."""

    def __init__(
        self,
        failure_threshold: int = 5,
        timeout: int = 60,
        expected_exception: type = Exception
    ):
        self.failure_threshold = failure_threshold
        self.timeout = timeout
        self.expected_exception = expected_exception
```

```
        self.state = CircuitState.CLOSED
        self.failure_count = 0
        self.last_failure_time = None

    async def call(self, func, *args, **kwargs):
        """Execute function with circuit breaker protection"""

        if self.state == CircuitState.OPEN:
            if time.time() - self.last_failure_time > self.timeout:
                self.state = CircuitState.HALF_OPEN
            else:
                raise CircuitBreakerError("Circuit is OPEN")

        try:
            result = await func(*args, **kwargs)
            self._on_success()
            return result
        except self.expected_exception as e:
            self._on_failure()
            raise

    def _on_success(self):
        """Reset on successful call"""
        self.failure_count = 0
        self.state = CircuitState.CLOSED

    def _on_failure(self):
        """Track failure and potentially open circuit"""
        self.failure_count += 1
        self.last_failure_time = time.time()

        if self.failure_count >= self.failure_threshold:
            self.state = CircuitState.OPEN
            logger.error(f"Circuit breaker OPENED after
{self.failure_count} failures")
```

---

## 10. Performance Metrics

### 10.1 Prometheus Metrics

```
# src/monitoring/metrics.py

from prometheus_client import Counter, Histogram, Gauge

# Request metrics
```

```
planning_requests_total = Counter(
    'planning_requests_total',
    'Total planning requests',
    ['status']
)

planning_latency = Histogram(
    'planning_latency_seconds',
    'Planning pipeline latency',
    ['stage'],
    buckets=[0.1, 0.5, 1.0, 2.0, 5.0, 10.0, 30.0]
)

# LLM metrics
llm_calls_total = Counter(
    'planning_llm_calls_total',
    'Total LLM API calls',
    ['model', 'stage']
)

llm_cost_usd = Counter(
    'planning_llm_cost_usd_total',
    'Total LLM cost in USD',
    ['model']
)

# Agent resolution metrics
agent_search_latency = Histogram(
    'agent_search_latency_seconds',
    'Agent search latency',
    ['method'],
    buckets=[0.01, 0.05, 0.1, 0.2, 0.5, 1.0]
)

agents_considered = Histogram(
    'agents_considered',
    'Number of agents considered',
    ['stage'],
    buckets=[1, 5, 10, 50, 100, 500, 1000]
)

# Validation metrics
validation_failures = Counter(
    'validation_failures_total',
    'Validation failures',
    ['tier', 'reason']
)

# Database metrics
```

```
db_query_latency = Histogram(
    'db_query_latency_seconds',
    'Database query latency',
    ['query_type']
)
```

### 10.2 Performance Benchmarks

```
┌─────────────────────────────────────────────────────────────┐
│              BASELINE PERFORMANCE (100K agents)             │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│  Operation                 │  p50    │  p95    │  p99       │
│  ──────────────────────────┼─────────┼─────────┼────────────│
│  Simple query planning     │  1.2s   │  1.8s   │  2.5s      │
│  Complex query planning    │  2.1s   │  3.2s   │  4.8s      │
│  Manifest processing       │  450ms  │  680ms  │  920ms     │
│  Vector search (2K)        │  120ms  │  180ms  │  250ms     │
│  Full-text search          │  15ms   │  25ms   │  35ms      │
│  Cross-encoder rerank      │  95ms   │  140ms  │  180ms     │
└─────────────────────────────────────────────────────────────┘

┌─────────────────────────────────────────────────────────────┐
│                    SCALABILITY TESTING                      │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│  Agent Count │ Planning Latency (p95) │ Search Latency      │
│  ────────────┼────────────────────────┼─────────────────────│
│  10K         │  1.5s                  │  100ms              │
│  50K         │  2.2s                  │  140ms              │
│  100K        │  3.2s                  │  180ms              │
│  500K        │  4.8s                  │  280ms              │
│  1M          │  6.5s                  │  350ms              │
└─────────────────────────────────────────────────────────────┘

┌─────────────────────────────────────────────────────────────┐
│              COST ANALYSIS (1000 queries/day)               │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│  Component              │  Daily    │  Monthly              │
│  ───────────────────────┼───────────┼───────────────────────│
│  LLM API calls          │  $0.10    │  $3.00                │
│  Database               │  $0.83    │  $25.00               │
│  Compute (3 instances)  │  $3.00    │  $90.00               │
│  Kafka                  │  $1.00    │  $30.00               │
│  ───────────────────────┼───────────┼───────────────────────│
│  Total                  │  $4.93    │  $148.00              │
└─────────────────────────────────────────────────────────────┘
```

```
┌──────────────────────────────────────────────────────────────────┐
│                    OPTIMIZATION IMPACT                           │
├──────────────────────────────────────────────────────────────────┤
│                                                                  │
│  Metric               │  Before │  After  │  Improvement       │
│  ─────────────────────┼─────────┼─────────┼────────────────────│
│  LLM calls/query      │  35     │  1      │  -97%              │
│  Avg latency          │  27s    │  2.15s  │  -92%              │
│  Cost per query       │  $0.070 │  $0.0001│  -99.8%            │
│  Agent resolution     │  15s    │  0.2s   │  -98.7%            │
└──────────────────────────────────────────────────────────────────┘
```

---

## 11. Testing Strategy

### 11.1 Test Pyramid

```
          ┌─────────────┐
          │   E2E (5%)  │  Full workflow tests
          └─────────────┘
        ┌─────────────────┐
        │ Integration (25%) │  Pipeline + DB tests
        └─────────────────┘
    ┌─────────────────────────┐
    │    Unit Tests (70%)     │  Component tests
    └─────────────────────────┘
```

### 11.2 Unit Test Examples

```python
# tests/unit/test_template_generator.py

import pytest
from src.manifest_processing.template_generator import
SemanticTemplateGenerator

@pytest.fixture
def sample_manifest():
    return {
        "identity": {
            "name": "TestAgent",
            "description": "Test agent",
            "tags": ["test", "mock"]
        },
        "capabilities": [
            {
                "type": "TOOL",
```

```
                "name": "test_tool",
                "description": "A test tool",
                "input_schema": {
                    "properties": {"input": {"type": "string"}}
                }
            }
        ]
    }

def test_generate_semantic_string(sample_manifest):
    """Test semantic string generation"""

    generator = SemanticTemplateGenerator()
    result = generator.generate(sample_manifest)

    assert "Agent: TestAgent" in result
    assert "Role: Test agent" in result
    assert "Expertise: test, mock" in result
    assert "TOOL named test_tool" in result

# tests/unit/test_decomposer.py

import pytest
from src.planning.decomposition.single_pass_decomposer import
SinglePassDecomposer

@pytest.mark.asyncio
async def test_simple_query_detection():
    """Test that simple queries are detected correctly"""

    decomposer = SinglePassDecomposer(mock_llm)

    simple_queries = [
        "Get ETH price",
        "What is the weather",
        "Convert 100 USD to EUR"
    ]

    for query in simple_queries:
        assert decomposer._is_simple_query(query) == True

@pytest.mark.asyncio
async def test_complex_query_detection():
    """Test that complex queries are detected correctly"""

    decomposer = SinglePassDecomposer(mock_llm)

    complex_queries = [
        "Get ETH price and alert if below $2000",
```

```
        "Monitor price every minute",
        "If price drops then execute trade"
    ]

    for query in complex_queries:
        assert decomposer._is_simple_query(query) == False
```

### 11.3 Integration Test Example

```
# tests/integration/test_planning_pipeline.py

import pytest
from src.planning.pipeline import OptimizedPlanningPipeline

@pytest.mark.asyncio
async def test_complete_planning_pipeline(test_db, mock_llm):
    """Test complete pipeline from query to manifest"""

    pipeline = OptimizedPlanningPipeline(test_db, mock_llm)

    query = "Monitor ETH price on Base. Alert if drops below $2800."

    # Execute pipeline
    manifest = await pipeline.plan(query)

    # Verify structure
    assert manifest.validated == True
    assert len(manifest.nodes) >= 2  # At least fetch + router
    assert any(n.type == "standard" for n in manifest.nodes)
    assert any(n.type == "router" for n in manifest.nodes)

    # Verify metrics
    assert manifest.planning_metrics['total_llm_calls'] <= 2
    assert manifest.planning_metrics['total_cost_usd'] < 0.001
```

---

## Document Metadata

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**Authors:** MetaOrcha Engineering Team
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**Related Documents:**

- Universal Agent Manifest Schema
- Runtime Service Specification
- Registry Service Specification
- Technical Architecture

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**END OF SPECIFICATION**